ESTADÍSTICA Y DISEÑO EXPERIMENTAL

LINEAR REGRESSION AND GLMs

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INTRODUCTION

The World is Encoded in Signals

Everything we observe — in business, engineering, biology, medicine, environment, society — is a **signal.**

But these signals are **noisy**: unrelated, mixed with randomness, measurement errors

Goal of science & statistics

Find patterns

To separate the **“true” underlying structure** from noise.

Turn data into understand the world

Linear models and generalized linear models (GLMs)

Describe the **relationship** between variables (explanatory models).

Predict **future outcomes** (predictive models).

FROM CORRELATION TO REGRESSION

Correlation: connection between two quantities. if a variable change, then another variable will change whether it could be directly or indirectly

Coefficients of correlation: Pearson, Spearman rank, and Kendall Tau

Pearson:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \ \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Spearman rank:

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} \qquad d_i = R(x_i) - R(y_i).$$

Spearman's ρ <u>is less sensitive to extreme outliers</u> than Pearson's r: uses ranks instead of raw values.

Spearman captures monotonic relationships (linear or non-linear).

FROM CORRELATION TO REGRESSION

Spearman rank:

Person	Weight (kg)	Height (cm)	Rank(Weight)	Rank(Height)	($d = r_x - r_y$)	(d^2)
P1	62	170	2	2	0	0
P2	85	182	6	8	−2	4
P3	73	176	4	4	0	0
P4	90	178	7	6	1	1
P5	55	165	1	1	0	0
P6	68	174	3	3	0	0
P7	80	180	5	7	−2	4
P8	95	177	8	5	3	9

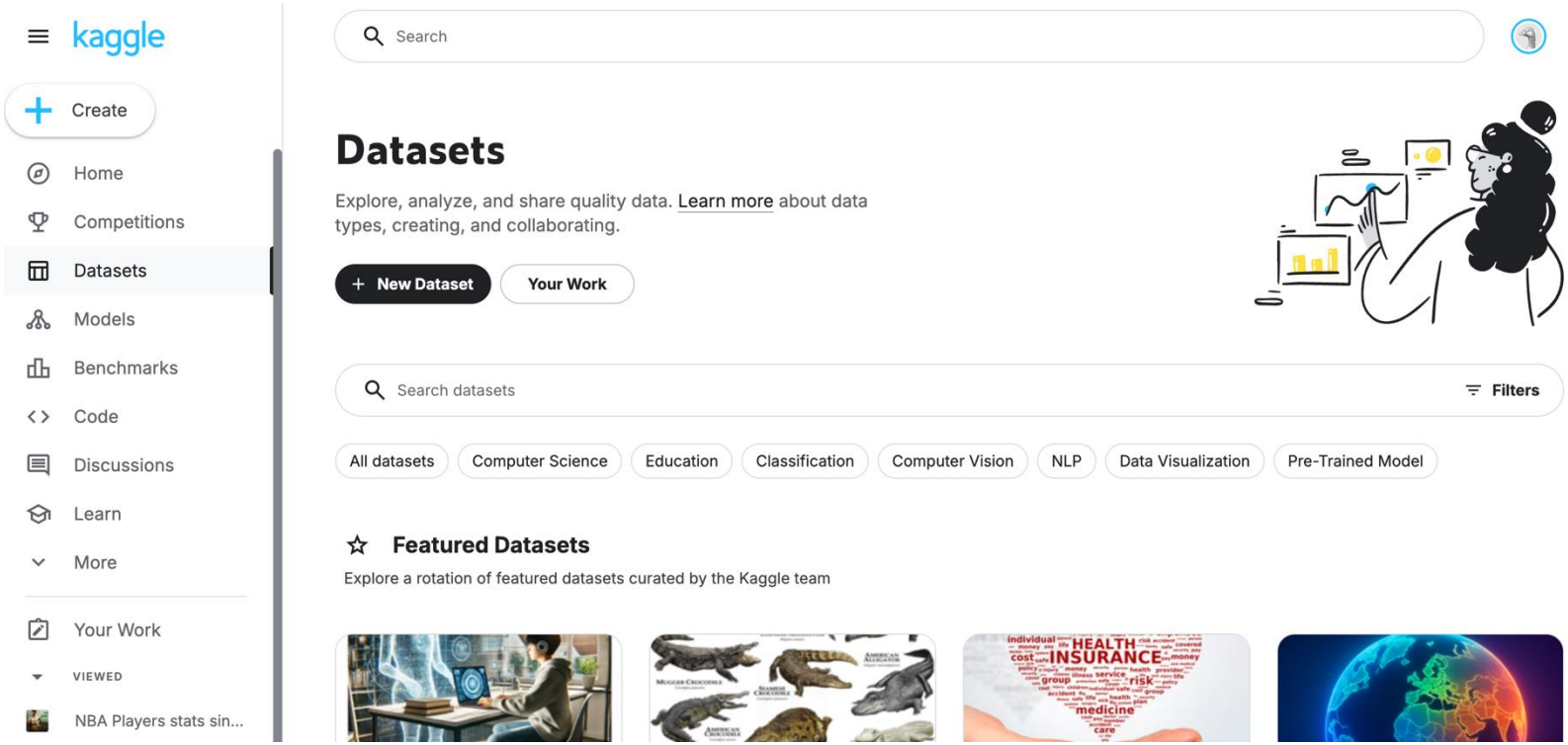
$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

n=8

$$\sum d^2 = 18$$

FROM CORRELATION TO REGRESSION

Exercise 1. NBA stats



FROM CORRELATION TO REGRESSION

Exercise 1. NBA stats – Minutes Played vs Points

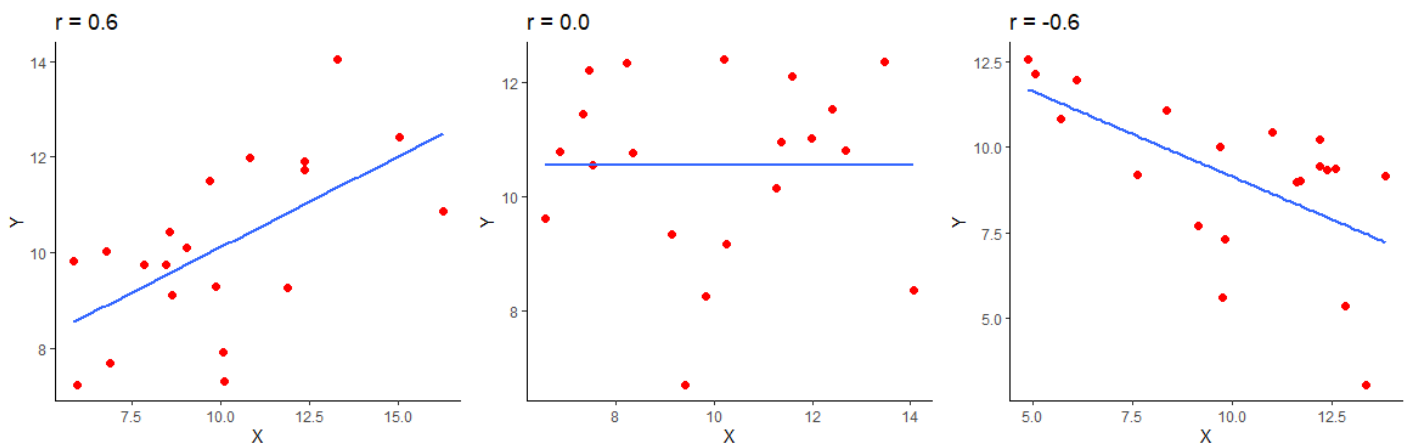
```
# https://www.kaggle.com/datasets/drgilermo/nba-players-stats?resource=download

# load data
season <- read_csv("archive/Seasons_Stats.csv")
nba_2017 <- season %>%
  dplyr::filter(Year == 2017) %>%
  dplyr::select(Player, MP, PTS) %>%
  dplyr::filter(!is.na(MP), !is.na(PTS), MP > 0, PTS > 0)

ggplot(nba_2017, aes(x = MP, y = PTS)) +
  geom_point(alpha = 0.6, color = "steelblue") +
labs(
    title = "Minutes Played vs Points Scored (NBA 2017 Season)",
    x = "Minutes Played",
    y = "Points"
  ) +
  theme_minimal()
```

FROM CORRELATION TO REGRESSION

Exercise 1. NBA stats

```
# 3. Compute correlations
pearson  <- cor(nba_2017$MP, nba_2017$PTS, method = "pearson")
spearman <- cor(nba_2017$MP, nba_2017$PTS, method = "spearman")

pearson; spearman

cor.test(nba_2017$MP, nba_2017$PTS, method = "pearson")
cor.test(nba_2017$MP, nba_2017$PTS, method = "spearman")
```

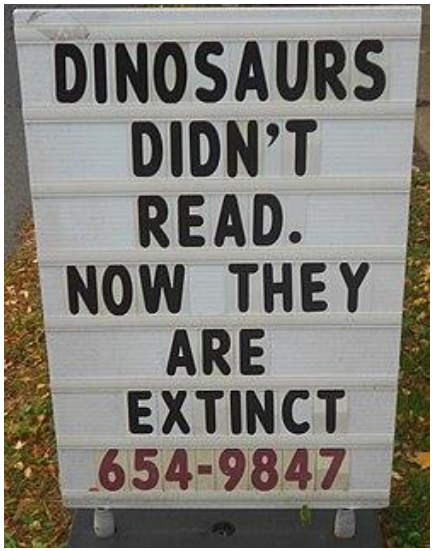
Meaning of cor.test



CORRELATION DOES NOT IMPLY CAUSATION

Possible alternative explanations

1. **Reverse causation** — the effect influences the supposed cause.
2. **Confounding / third variable** — a hidden factor causes both variables.
3. **Bidirectional causation** — each variable affects the other.
4. **Coincidence / spurious relationship** — the correlation is by chance.



Correlations are useful for hypothesis generation and prediction

REGRESSION - LINEAR MODELS

- Regression quantifies the relationship.
- Regression tells us: “for each additional cm of height, weight increases by X kg on average.”

Mathematical representation:

$$Y = \alpha + \beta X + \varepsilon$$

Components:

<u>Response variable</u> (Y) – what we want to explain.

<u>Intercept</u> (α) – baseline when X = 0.

<u>Explained part</u> (βX) – explained part

<u>Error term</u> (ε) – unexplained part: accounts for variability not explained by predictors.

REGRESSION - LINEAR MODELS

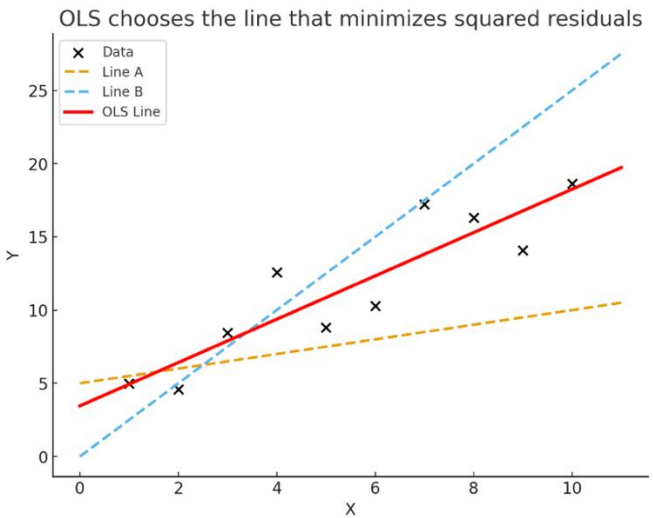
Minimize the sum of the squared differences between the observed and predicted values.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2$$

Ordinary least squares (OLS)

$$\hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

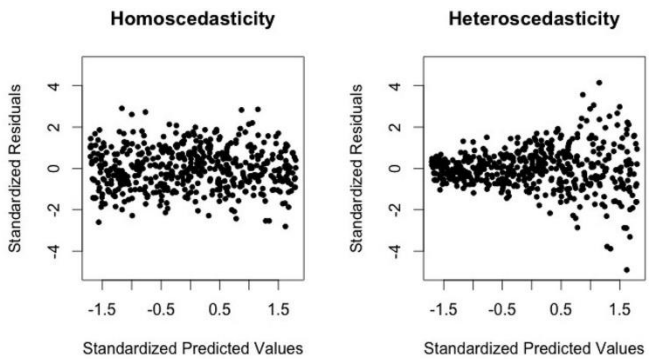


lm() function in R runs Ordinary Least Squares (OLS)

REGRESSION - LINEAR MODELS

Assumptions

1. Linear relationship between predictors and outcome
   - Plot the data!
2. Independent observations
   - If you measure the same person's blood pressure 5 times, those 5 values are correlated.
3. Constant variance of errors (homoscedasticity)
4. Normally distributed errors
5. No strong multicollinearity among predictors



REGRESSION - LINEAR MODELS

 # Exercise 2. Car Weight vs Fuel Efficiency

https://www.kaggle.com/datasets/uciml/autompg-dataset

```
# install.packages(c("tidyverse","broom","ggpubr","lmtest","car"))
library(tidyverse)
library(broom)
library(ggpubr)
library(lmtest)   # bptest, dwtest

# Load dataset -------------------------------------------------------
auto <- read_csv("auto-mpg.csv")  # file from Kaggle
names(auto)

df <- auto %>%
  dplyr::select(mpg, weight) %>%
  dplyr::filter(is.finite(mpg), is.finite(weight))

## try with and without log2
df$mpg <- log2(df$mpg)
```

REGRESSION - LINEAR MODELS

 # Exercise 3. BMI vs Medical charges

https://www.kaggle.com/datasets/mirichoi0218/insurance

```
# Load and clean -------------------------------------------------------
insurance <- read_csv("insurance.csv", show_col_types = FALSE)

df <- subset(insurance)  %>%
  dplyr::select(charges, bmi) %>%
  dplyr::filter(is.finite(charges), is.finite(bmi))

ggplot(df, aes(bmi, charges)) +
  geom_point(alpha = 0.35) +
  labs(title = "Charges vs BMI", x = "BMI", y = "Medical Charges") +
  theme_minimal()

## any problem?
# check other colums!
```

REGRESSION –MULTIPLE LINEAR MODELS

- Multiple predictors.
- Assumes the response y is continuous, normally distributed (for inference), with constant variance.

$$y_i = \alpha + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i$$

- y_i = outcome
- $x_{i1}, x_{i2}, \ldots, x_{ip}$ = predictors (age, weight, income, etc.)
- $\beta_1, \beta_2, \ldots, \beta_p$ = coefficients for each predictor
- α = intercept
- ε_i = error

REGRESSION – MULTIPLE LINEAR MODELS

Exercise 4. Updated BMI vs Medical charges

https://www.kaggle.com/datasets/vipullrathod/fish-market

```
# Load data
ins <- read_csv("insurance.csv")

# Convert categorical variables
ins <- ins %>%
  mutate(
    smoker = factor(smoker),
    sex    = factor(sex)
  )

# Single predictors
m1 <- lm(charges ~ bmi, data=ins)
m2 <- lm(charges ~ age, data=ins)
m3 <- lm(charges ~ smoker, data=ins)

# Multiple predictors
m4 <- lm(charges ~ age + bmi + smoker, data=ins)
```

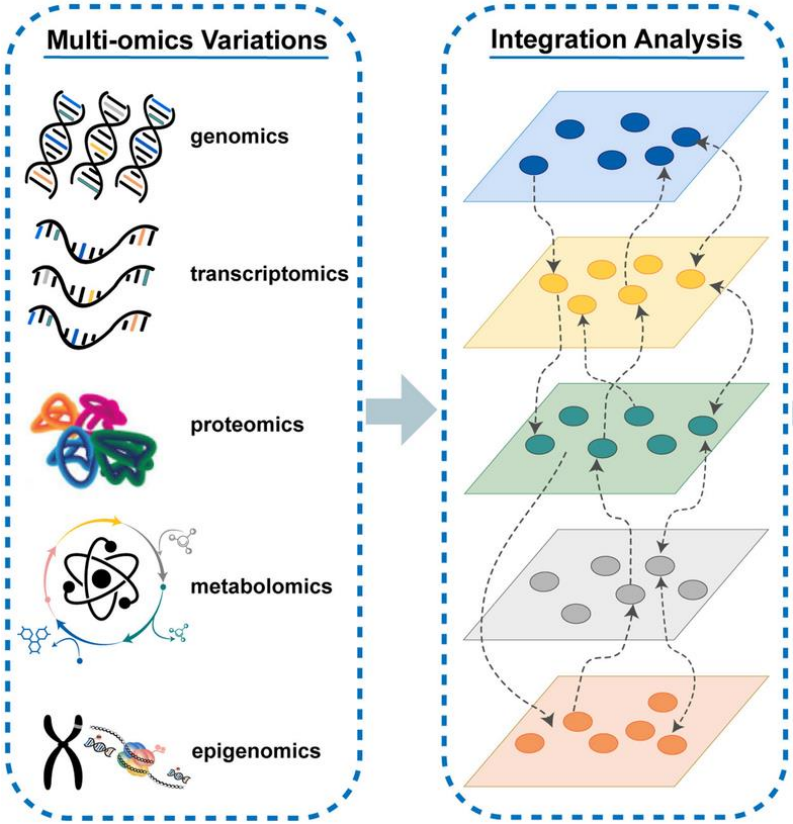
REGRESSION – MULTIPLE LINEAR MODELS

 # Exercise 4. Updated BMI vs Medical charges

https://www.kaggle.com/datasets/vipullrathod/fish-market

```
# Compare adjusted R²
glance(m1)$adj.r.squared
glance(m2)$adj.r.squared
glance(m3)$adj.r.squared
glance(m4)$adj.r.squared

# Summaries
summary(m4)
```

REGRESSION – SCREENING APPROACHES



When we test many predictors, some will look significant just by chance.

Reason: If you test 100 predictors at α = 0.05, you expect ~5 false positives even if none are truly related.

FDR correction controls the expected proportion of false positives among the significant results.

REGRESSION – SCREENING APPROACHES

 # Exercise 5 – FDR correction

https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

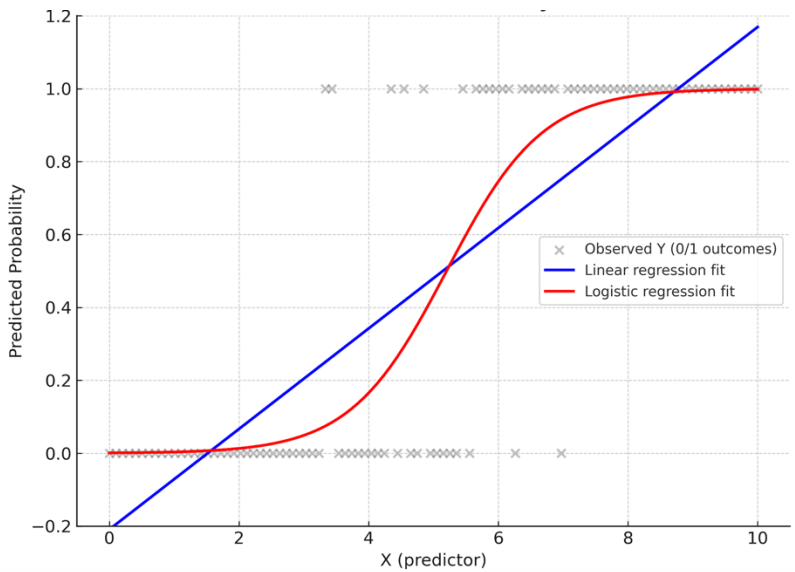
```
library(tidyverse)
library(broom)

# Load data (train.csv from Kaggle)
house <- read_csv("train.csv")

# Keep only numeric predictors
num_vars <- house %>%
  dplyr::select(where(is.numeric)) %>%
  drop_na()

names(num_vars) <- make.names(names(num_vars))

# Outcome
outcome <- "SalePrice"
predictors <- setdiff(names(num_vars), outcome)

...
```

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

Linear models: outcome must be continuous, errors ~ Normal

But many outcomes are **not continuous**:

1. **Binary** (yes/no disease)
2. **Counts** (number of hospital visits)
3. **Rates** (defects per product, survival time). a linear model could predict impossible values (like –0.2 or 1.3).

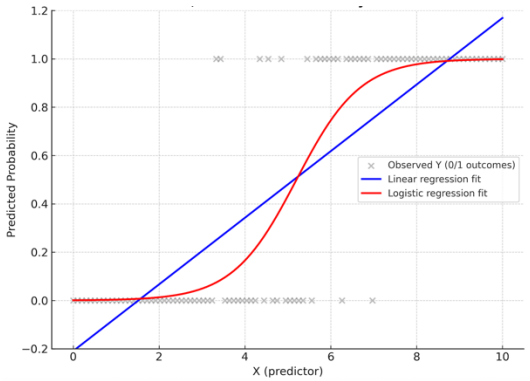


FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

Linear models are:

$$Y = X\beta + \varepsilon$$

Errors: $\varepsilon \sim N(0, \sigma^2)$ are explicit.



GLMs extend linear models by (a more flexible definition):

$$g(\mu) = X\beta$$

$$\mu = E[Y|X]$$

$\boldsymbol{\mu} = \boldsymbol{E}[\boldsymbol{Y}|\boldsymbol{X}]$ is the expected value of outcome given an X.

We don't model the raw outcome Y directly (too noisy and restrictive).

Instead, we model the expected value of Y given X.

For every value of the predictors X (e. BMI=30), there is a distribution of possible outcomes Y

1. *That distribution is a random component (Normal, Binomial, Poisson, ...).*
2. *The mean of that distribution is μ=E[Y/X].*

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

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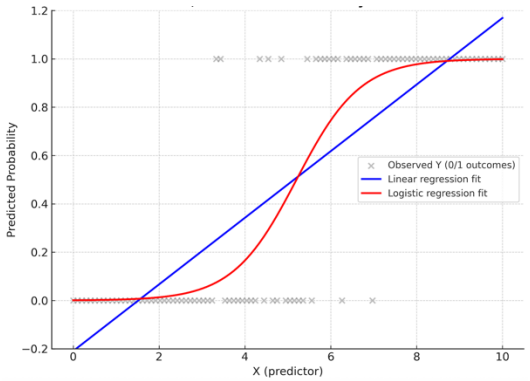

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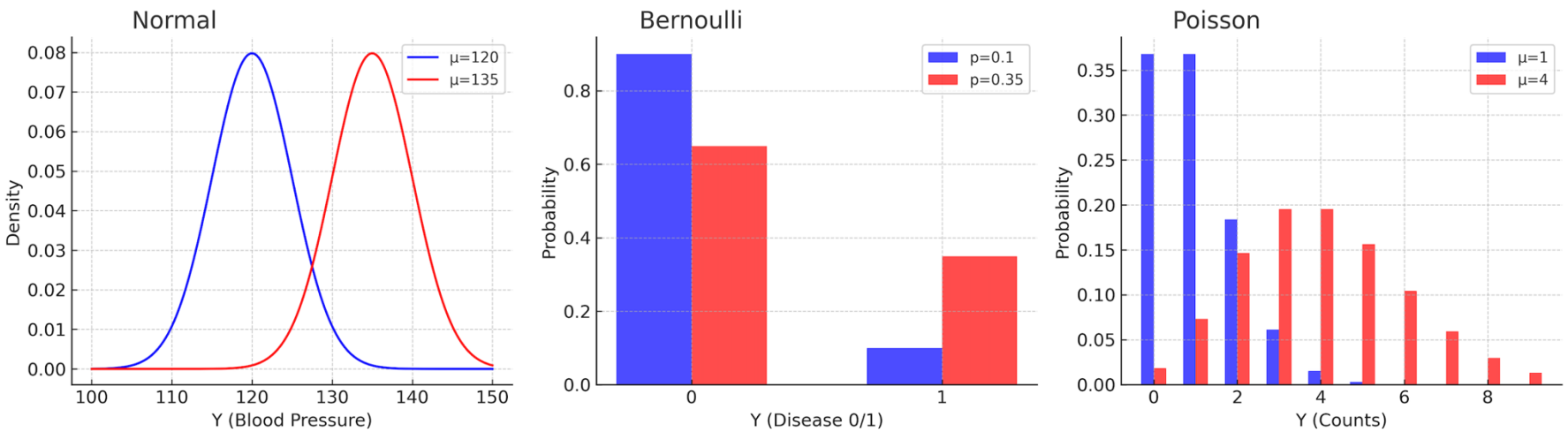

Red - Blue; different X values

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

Linear models are:

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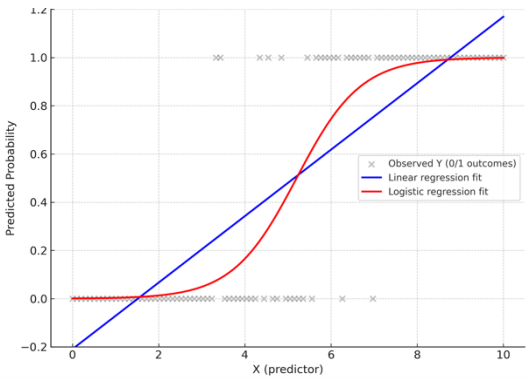


GLMs extend linear models by (a more flexible definition):

$$g(\mu) = X\beta$$
$$\mu = E[Y|X]$$

$g(\cdot)$= **link function**

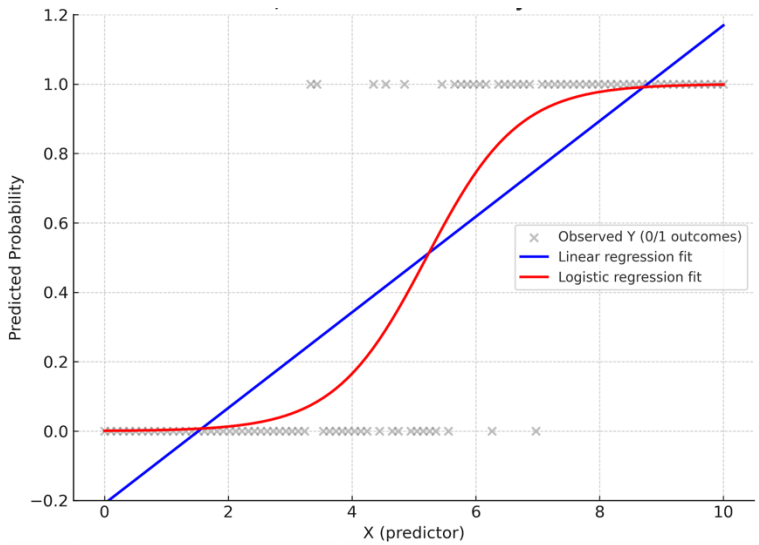
- The **link function** transforms the expected value into a space where a **linear relationship with predictors makes sense**.

Identity link: $g(\mu) = \mu$; Linear regression.

Logit link: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$; Logistic regression

Log link: $g(\mu) = \log(\mu)$; Poisson regression

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

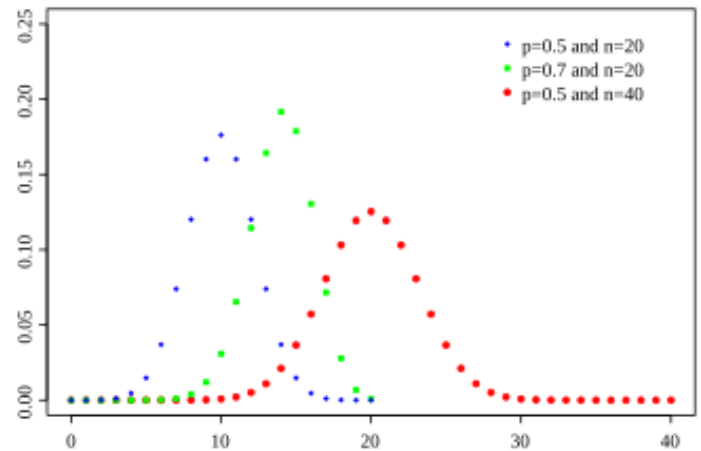



```
glm_fit <- glm(y ~ x, data = df, family = binomial(link = "logit"))
```

Distribution = Binomial
Link = logit
It models probability of disease ∈ [0,1]

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

The Binomial distribution describes the number of successes in a fixed number of independent trials, where each trial has two outcomes (success or failure) and the same chance of success.



$$P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \; k = 0 1 2 \ldots n$$

n= number of trials
k= number of successes
p= probability of success in each trial

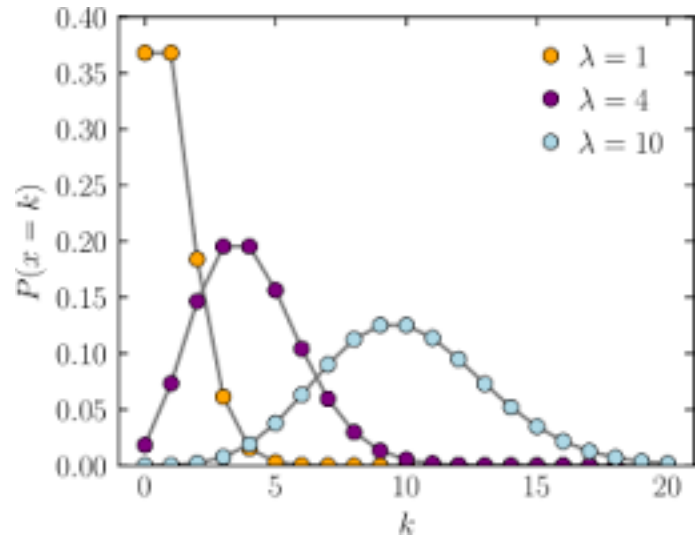
FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

```
# Exercise 6. GLMs
# BMI vs Heart Event (0/1) – Logistic regression
df <- data.frame(
  BMI = c(
    18.5, 19.2, 19.8, 20.1, 20.5, 21.0, 21.5, 22.0, 22.5, 23.0,
    23.5, 24.0, 24.5, 25.0, 25.5, 26.0, 26.5, 27.0, 27.5, 28.0,
    28.5, 29.0, 29.5, 30.0, 30.5, 31.0, 31.5, 32.0, 32.5, 33.0,
    33.5, 34.0, 34.5, 35.0, 35.5, 36.0, 36.5, 37.0, 37.5, 38.0,
    38.5, 39.0, 39.5, 40.0, 40.5, 41.0, 41.5, 42.0, 42.5, 43.0
  ),
  Event = c(
    0,0,0,0,0, 0,0,0,0,0,
    0,0,0,0,0,0,0,0,0,0,
    0,0,0,0,1,0,1,0,1,0,
    1,1,1,1,1,1,1,1,1,1,
    1,1,1,1,1,1,1,1,1,1
  )
)
```

Logit link: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$; Logistic regression

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

The Poisson distribution is used for counting the number of times an event occurs in a fixed interval of time or space when events happen at a constant average rate and are independent of each other.



$$P(Y = k) = \frac{e^{-\mu}\,\mu^k}{k!}, k = 0,1,2, \ldots$$

Y= count of events (random variable)

k = actual observed count (0, 1, 2, ...) | Number of Hospital visits/year

μ= expected number of events in that interval (the mean)

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

```
# Exercise 7. Visits vs BMI / Poisson
df <- data.frame(
  BMI = c(
    18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
    28, 29, 30, 31, 32, 33, 34, 35, 36, 37,
    38, 39, 40
  ),
  Visits = c(
    0, 0, 0, 0, 0, 0, 1, 0, 1, 1,
    1, 1, 2, 2, 2, 3, 3, 4, 4, 5,
    6, 7, 8 )
)

ggplot(df, aes(BMI, Visits)) +
  geom_point(alpha = 0.7) +
  labs(title = "BMI vs Number of Hospital Visits",
       subtitle = "Points = observed counts\n",
       y = "Number of Visits",
       x = "BMI") + theme_minimal()
```

What would you do?

FROM LINEAR MODELS TO GENERALIZED LINEAR MODELS (GLMS)

 # Exercise 8. BMI vs Charges

```
# BMI vs Charges
df <- data.frame(
  BMI = c(18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40),
  Charges = c(2100, 2300, 2600, 3100, 3400,
            4200, 4800, 6700, 7600, 10300,
            12800, 15000))

# Scatter plot
ggplot(df, aes(BMI, Charges)) +
  geom_point(size = 2, alpha = 0.7) +
  labs(title = "BMI vs Medical Charges",
      subtitle = "Positive, right-skewed with noise",
      x = "BMI", y = "Charges (USD)") +
  theme_minimal()
```

What would you do?